Embedded Systems and IoT

Ingegneria e Scienze Informatiche - UNIBO
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Lecturer: Prof. Alessandro Ricci

[module 3.2] FROM EMBEDDED SYSTEMS TO INTERNET OF THINGS

OUTLINE

- Internet of Things - Introduction
- From IoT to Enterprise/Industrial IoT
- IoT and Web of Things

INTERNET OF THINGS

- **Internet of Things** (IoT)
  - the name was first introduced in 1999 by Kevin Ashton, as founder of Auto-ID Center at MIT, about projects concerning RFID technologies
  - initial objective: *to automate the digitalisation of the physical world (objects, their id, measures, events..), by means of sensors to feed data at real-time to the Internet, avoiding the manual error-prone insertion by people* [*]
- Strong impact to the society [**]
  - not only at the technical level

[*] Kevin Ashton. "The 'Internet of Things' Thing'. RFID Journal, 2009. http://www.rfidjournal.com/articles/view?4986]

[**] Samuel Greengard. *The Internet of Things. MIT Press*

IoT: A DEFINITION

- A definition [LIT]:
  - *"The IoT is what we get when we connect Things, which are not operated by humans, to the Internet"*
    - systems composed by physical objects that are connected and communicate by means of the Internet
    - these physical objects are typically embedded systems
- IoT is at the core of **Industrial Internet**
  - infrastructure supporting large-scale and robust connectivity between devices and data
  - integration of embedded system with sensors, software and telecom systems
  - also called *"Industry 4.0" o smart industry o smart manufacturing*

INTERNET EVOLUTION

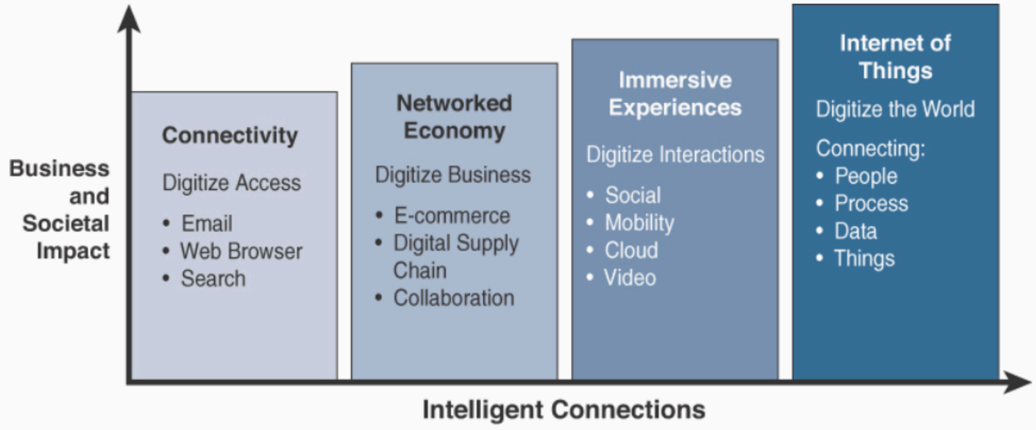


Figure 1-1 *Evolutionary Phases of the Internet*

[IoTF, p.68]

Internet Phase	Definition
Connectivity (Digitize access)	This phase connected people to email, web services, and search so that information is easily accessed.
Networked Economy (Digitize business)	This phase enabled e-commerce and supply chain enhancements along with collaborative engagement to drive increased efficiency in business processes.
Immersive Experiences (Digitize interactions)	This phase extended the Internet experience to encompass widespread video and social media while always being connected through mobility. More and more applications are moved into the cloud.
Internet of Things (Digitize the world)	This phase is adding connectivity to objects and machines in the world around us to enable new services and experiences. It is connecting the unconnected.

Table 1-1 *Evolutionary Phases of the Internet*

IoT - MAIN ELEMENTS

- "Things"
  - embedded systems
  - equipped with sensors, in particular
- **Connectivity and communication**
  - adoption of specific **communication protocols** on top of the Internet
    - e.g. MQTT, CoAP
  - **interoperability** issue
  - IoT as System-of-Systems
- Critical aspects
  - security
    - identity, authentication, authorisation
  - privacy & data ownership

SMART THINGS

- Simplified version [DIT]:

IoT =
Physical Objects +
Controller/Sensors/Actuators +
Internet

- …where the "things" (physical object) can be very different kind of objects that people use daily
- ***Enchanted objects*** *vision* - D. Rose
  - http://tedxtalks.ted.com/video/TEDxBerkeley-David-Rose-Enchant

SMART THINGS EXAMPLES

- WeMo light switch & family
  - http://www.belkin.com/us/p/P-F7C030/
- Kevo smart lock
  - http://www.kwikset.com/kevo/
- Fitbit force wristband
  - http://www.fitbit.com/
- Metromile
  - https://www.metromile.com/
- Google Nest thermostat & family
  - https://nest.com/
- …
- enchanted objects
  - http://enchantedobjects.com/products/

SMARTPHONE ROLE

- Smartphone role in IoT
  - can be themselves embedded systems
    - being equipped by many different kinds of sensors, to collect data, geo-localised, and send it through the Internet
  - can be a *universal "remote controller"*
    - uniform user interface (UI) to interact with smart things
  - can be the control unit governing near devices by means of protocols such as Bluetooth, possibly connected with other wearable devices
    - wrists, smartwatches, smartglasses

IoT EVOLUTION

- 5 stages [Por14,Por15,DBE17]
  1. **product stage**
     - *the air conditioner*
  2. **smart product**
     - *the programmable air conditioner*
  3. **smart connect products**
     - *air conditioner accessible by the Internet*
       - controllable by the phone
       - the company can check its functioning and compare to other millions of other units to do *predictive maintenance*

  …

IoT EVOLUTION

- 5 stages [Por14,Por15,DBE17] (cont.)
  4. product systems
  - *the smart thermostat talks to the connected HVAC and the smart window blinds and heated floors*
  - key points
    - **interoperability** & communication standards
      - » Web of Things (WoT) Initiative
    - **command and control platforms** availability
      - » Apple HomeKit, Amazon Echo, Google Home, Samsung SmartThings...
      - » industrial counter part: GE's Predix and Hitachi's Lumada

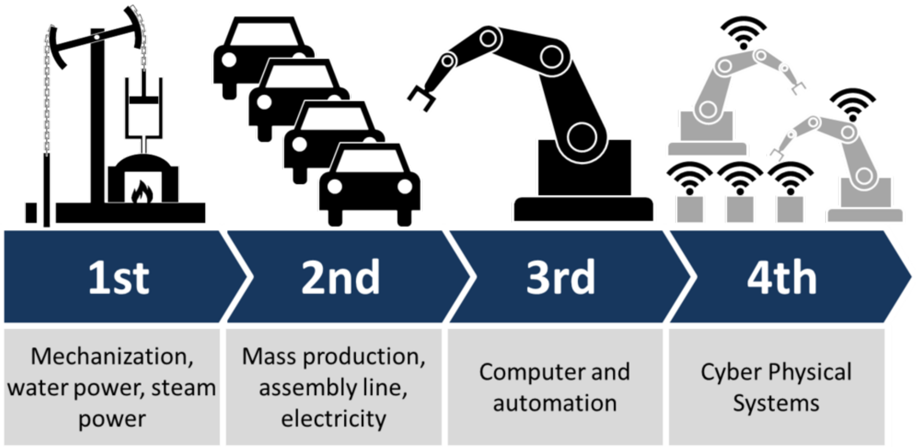
IoT EVOLUTION

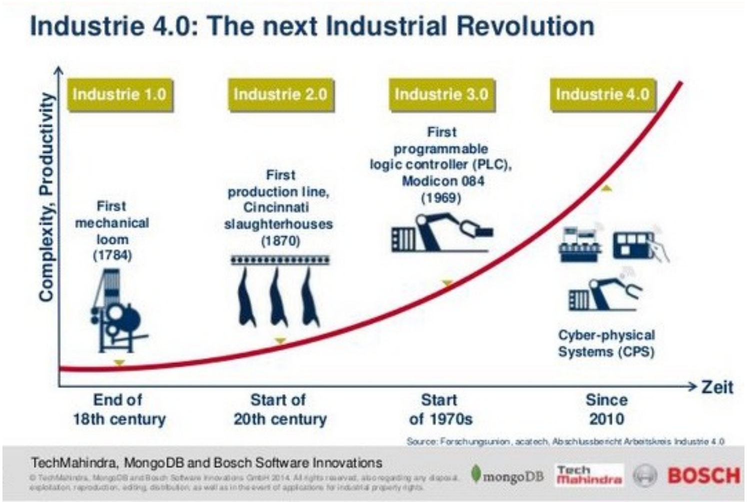
- 5 stages [Hep14,DBE17] (cont)
  5. system of systems
  - home appliances talking with home security talking to the car and wearable devices talking to smart hospital…
  - key points
    - interoperability
    - scalability, openness
    - governance

FROM IOT TO ENTERPRISE/INDUSTRIAL IOT

- IoT as key technology for industry & enterprises
  - **Industry 4.0**
    - fourth industrial revolution
  - key aspect: **data**
    - collect data at real-time through sensors
    - *big-data, real-time, big-stream*
- Enterprise IoT vs Industrial IoT (I-IoT)
  - **Enterprise IoT** => general term used to refer at IoT used at the enterprise level
  - **Industrial IoT (I-IoT)** => characterisation of Enterprise IoT specifically in the context of manufacturing & industries

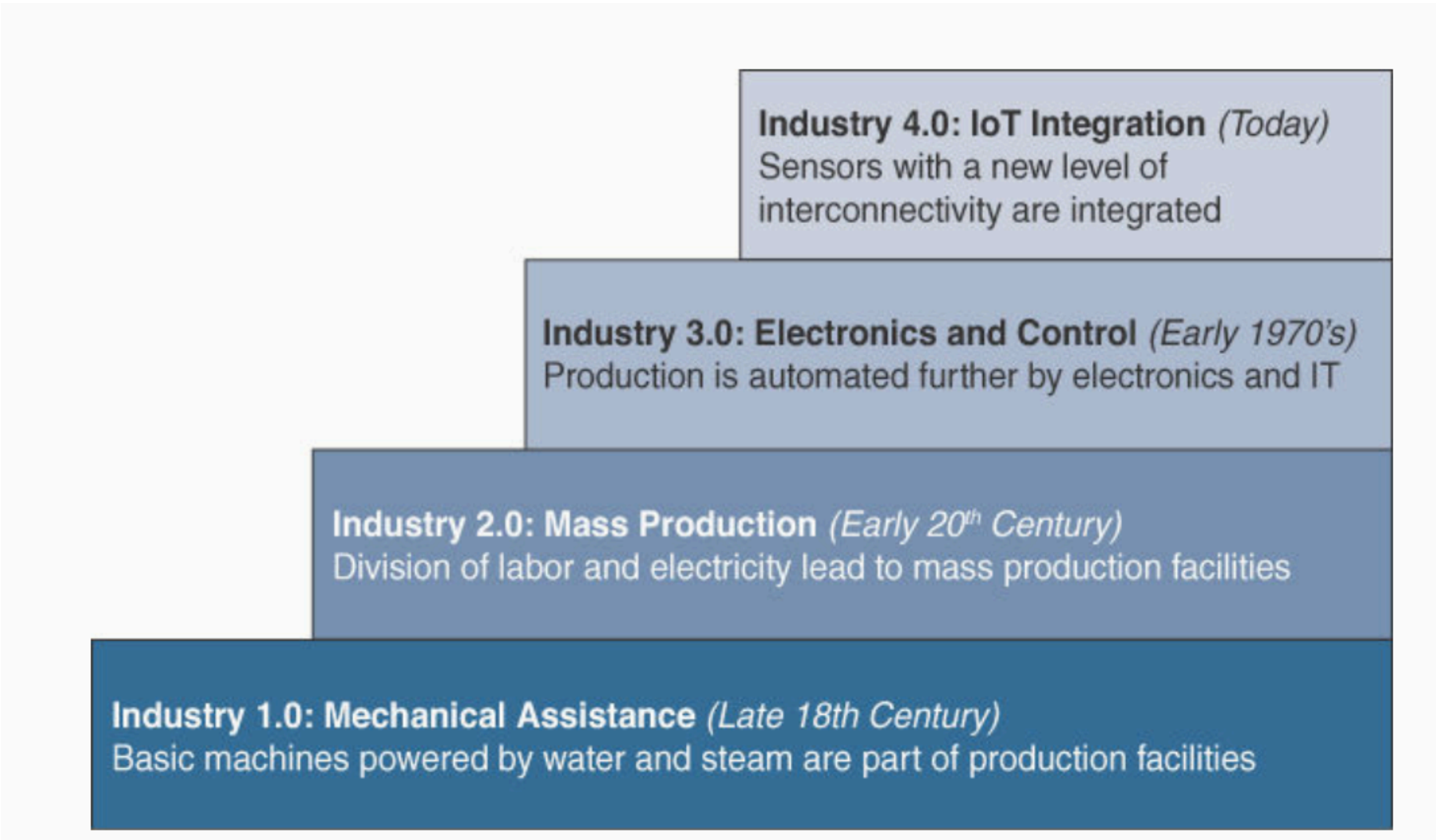
INDUSTRY 4.0





http://www.eesc.europa.eu/?i=portal.en.group-1-new-news.34501

FOUR INDUSTRIAL REVOLUTIONS



[IoTF, 85]

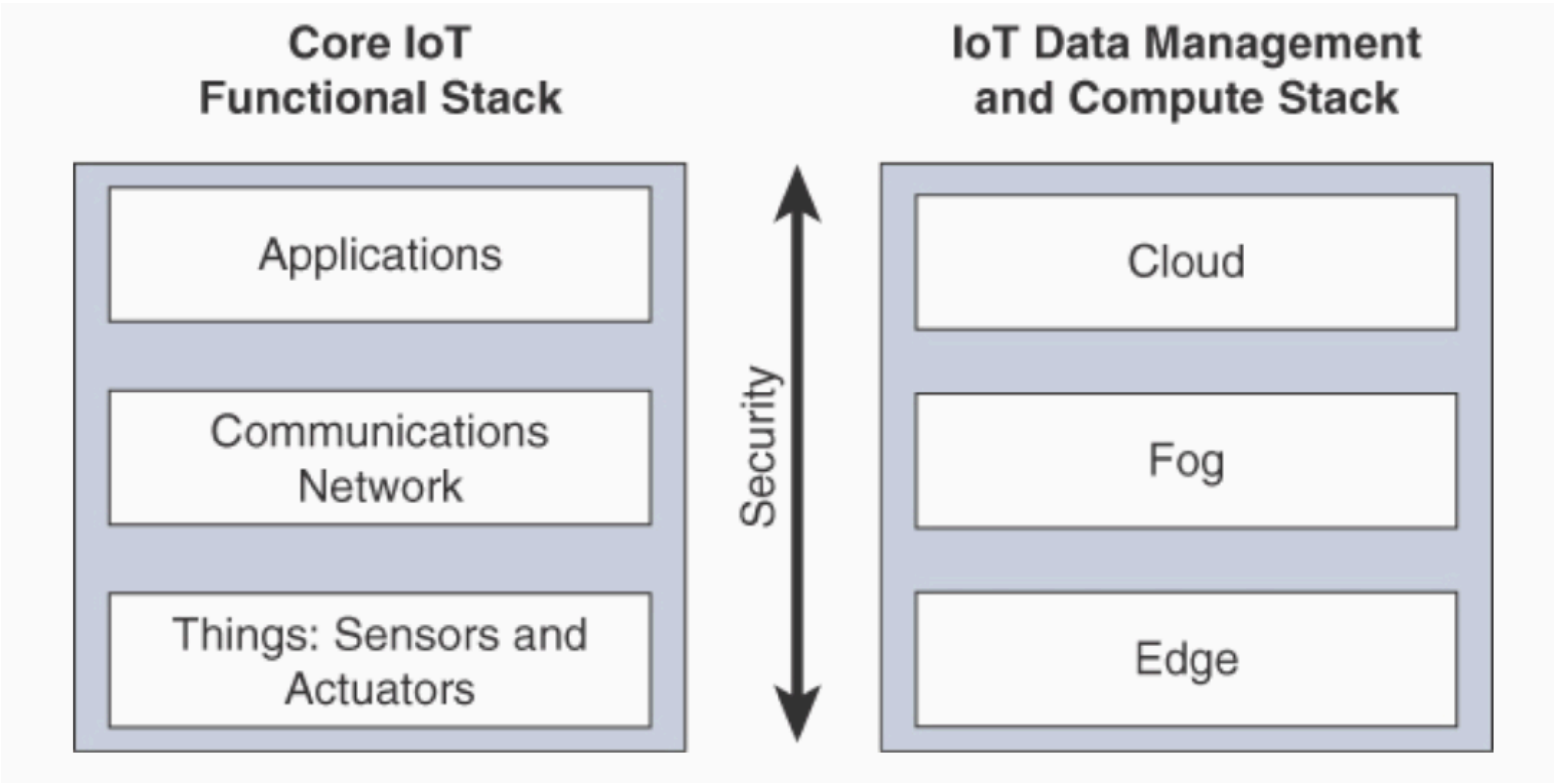
ENTERPRISE IOT

- Conceptual Framework to identify the Enterprise IoT level [*]
- Integration of IoT with Artificial Intelligence (AI) / Machine Learning (ML) often called as ***Precision <Something>***
  - Precision Medicine, Precision Agricolture, …

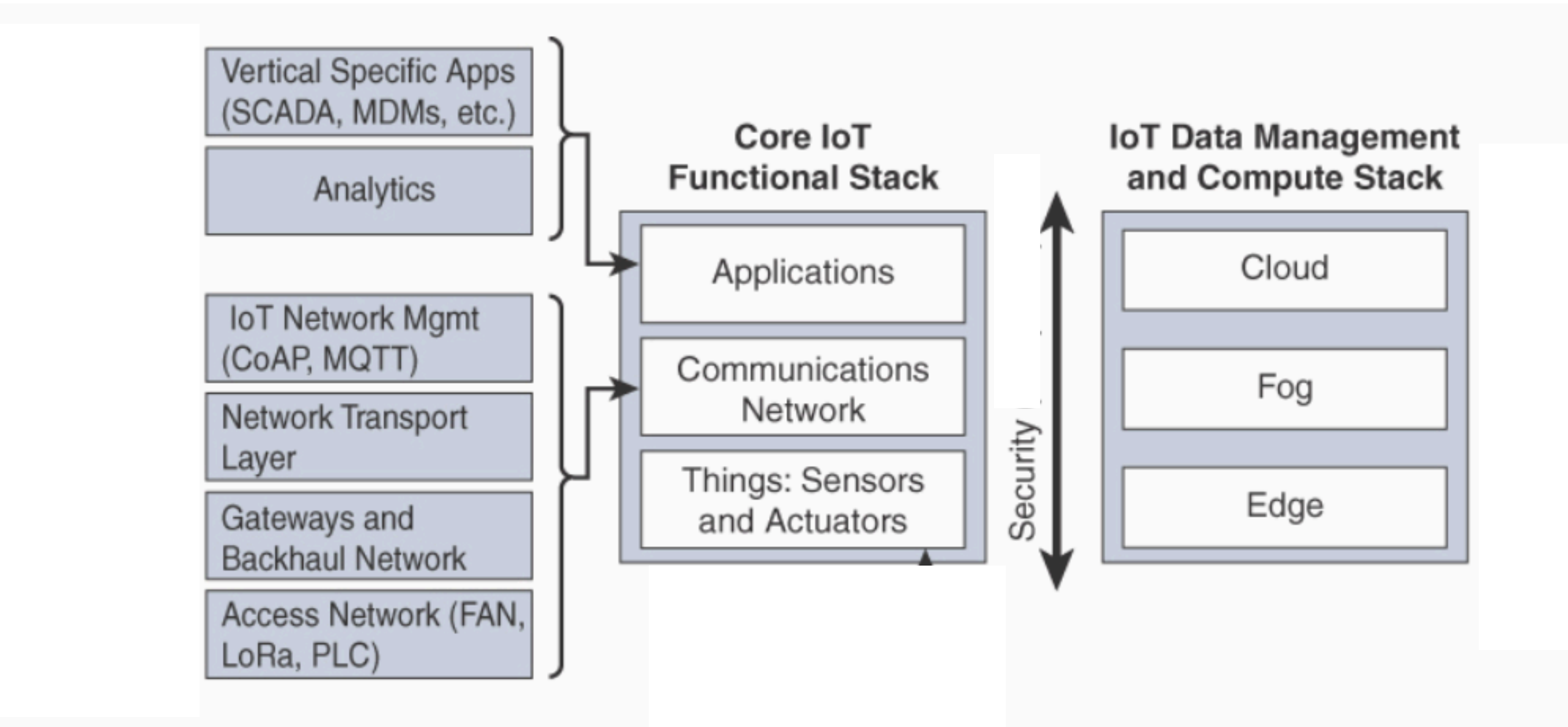
Do	• Enterprise applications driving business outcomes • Precision machine services
Learn	• technologies that allow to learn from data collected • learning and analysis systems including query technology and both supervised and unsupervised machine learning
Collect	• Technologies designed to collect data, which are increasingly time-series data being sent every hour, minute, second • BigData, BigStream technologies
Connect	• things can be connected to the Internet in various ways. • Choices depend on amount of data to be transmitted, how far you need to go, how much power you have, how fast data should arrive…
Things	• embedded systems & smart things • sensing capabilities in particular

(*) Timothy Chou. Precision. Principles, Practices and Solutions for the Internet of Things. Crowdstory, 2016

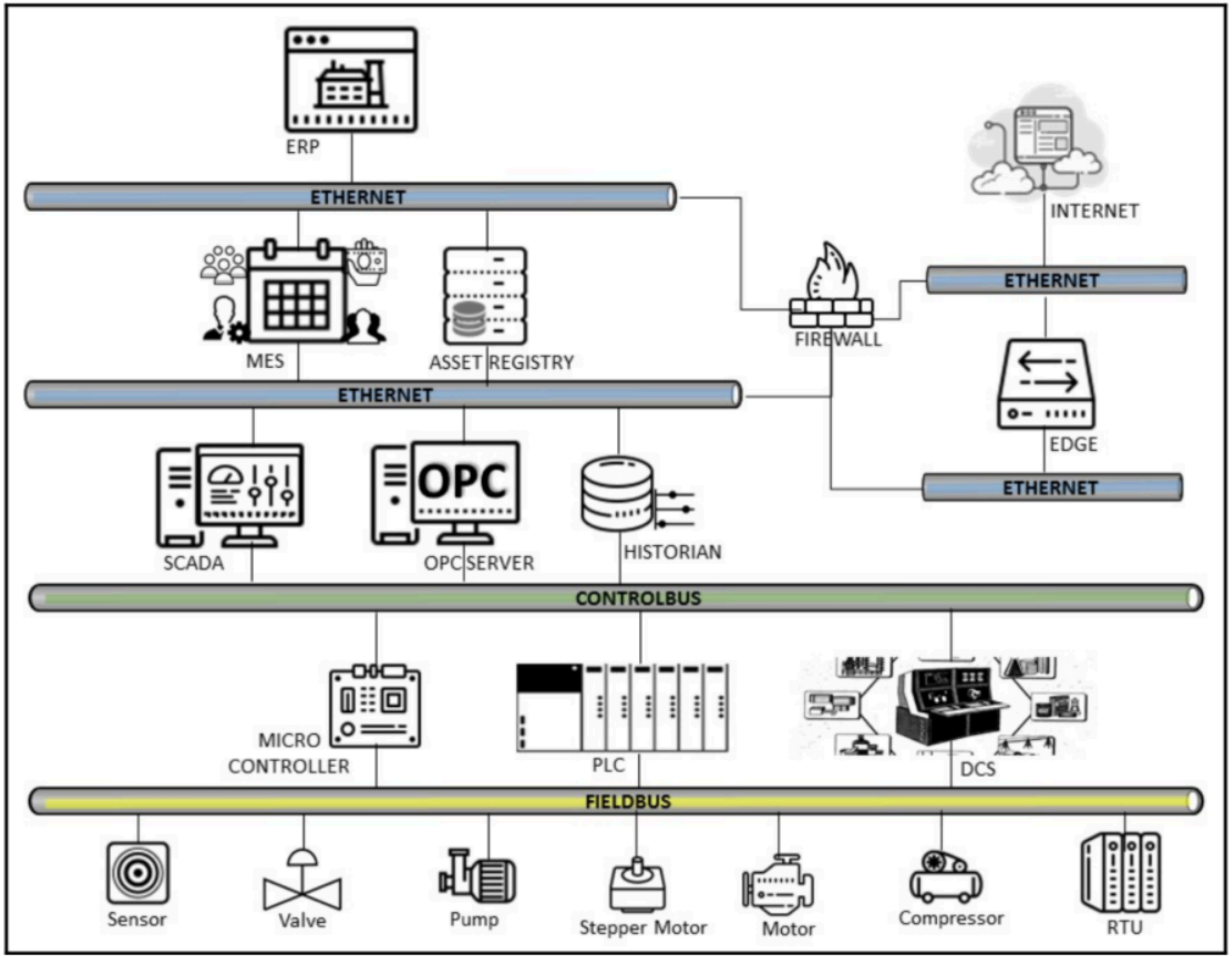
SIMPLIFIED ARCHITECTURE OF AN ENTERPRISE IOT SYSTEM



SIMPLIFIED ARCHITECTURE OF AN ENTERPRISE IOT SYSTEM



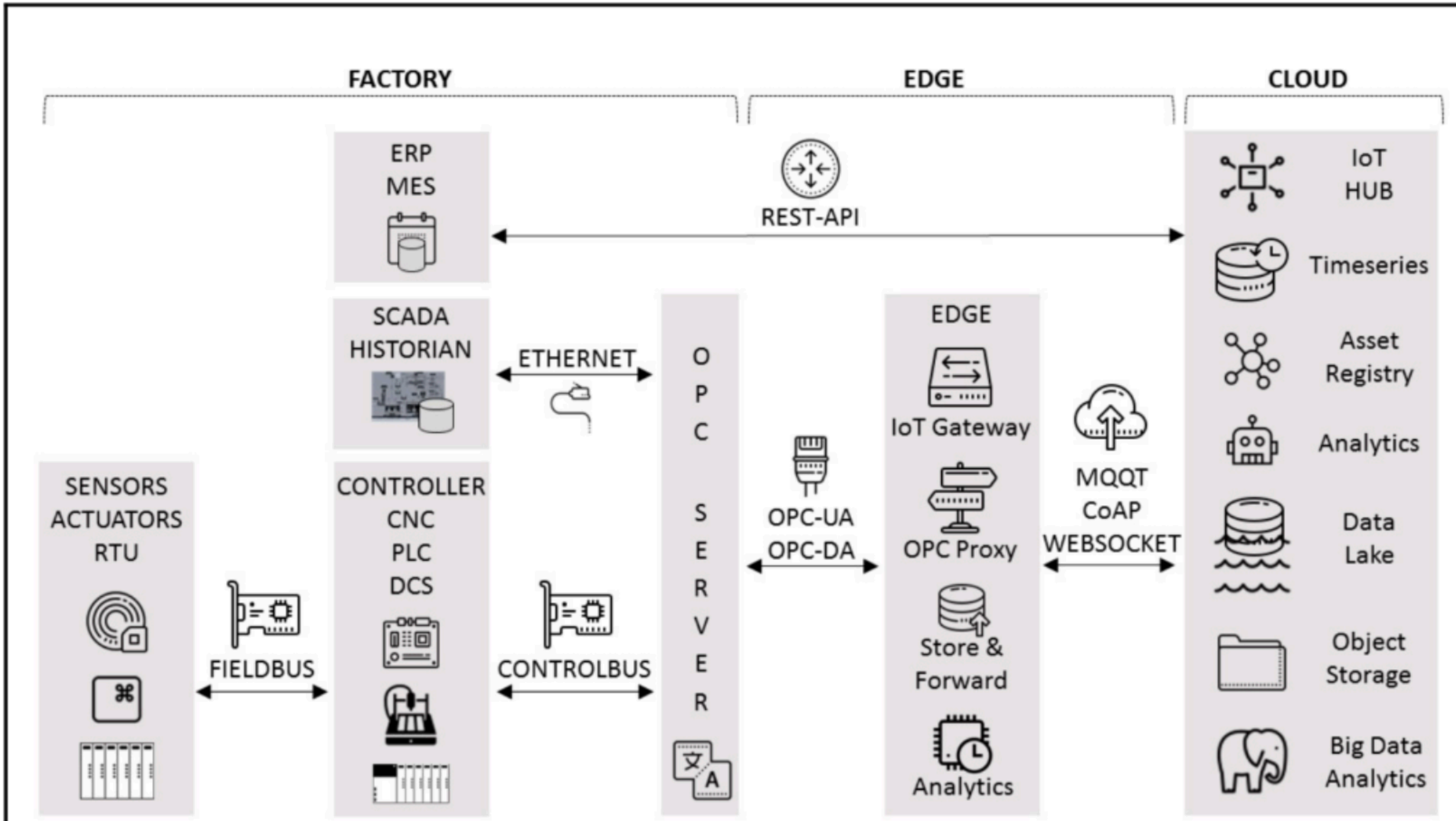
I-IOT DEVICES & PROTOCOLS



[VC18]

I-IOT DATA FLOW

- From data generation to their processing on the cloud



[VC18]

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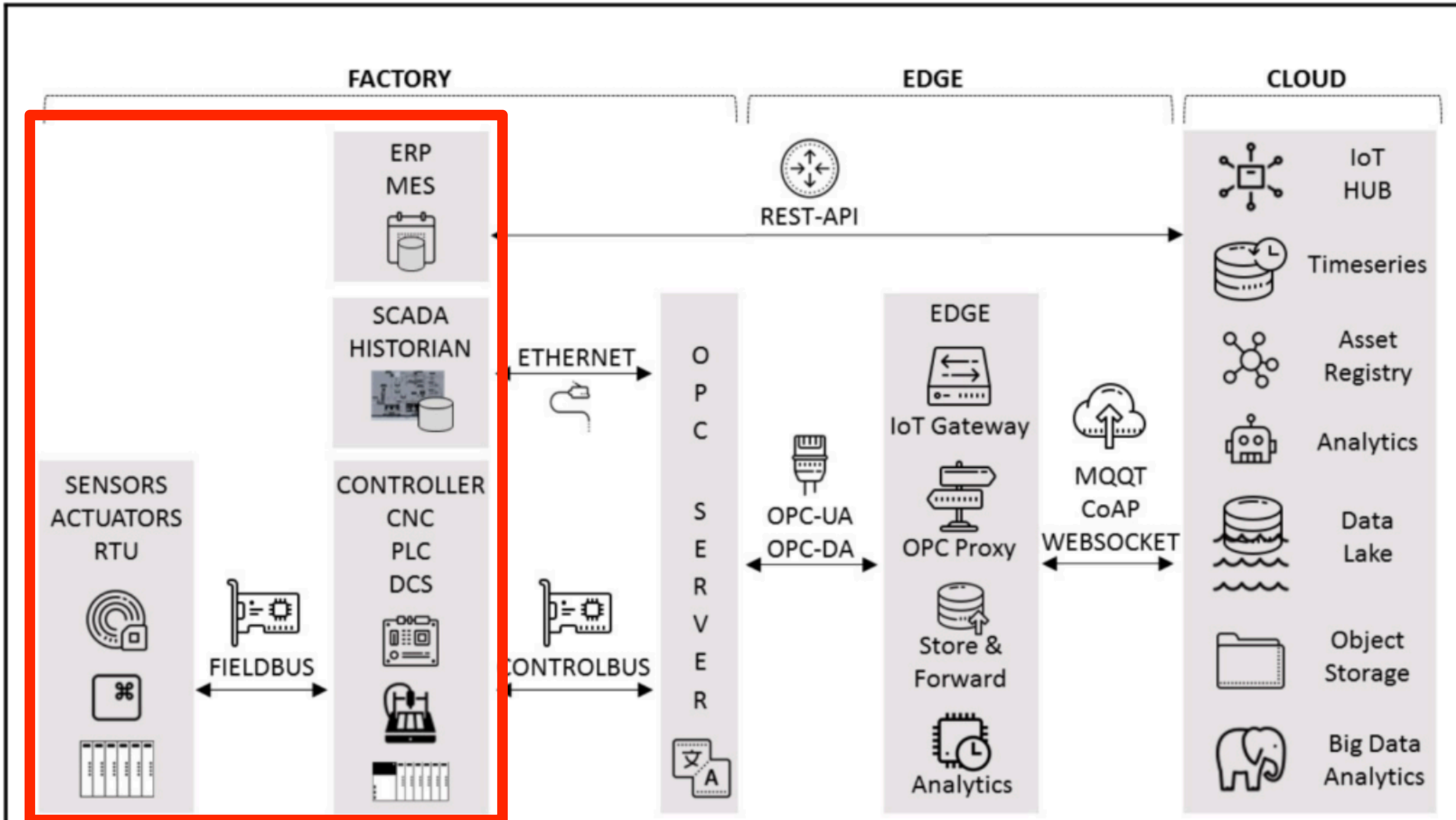


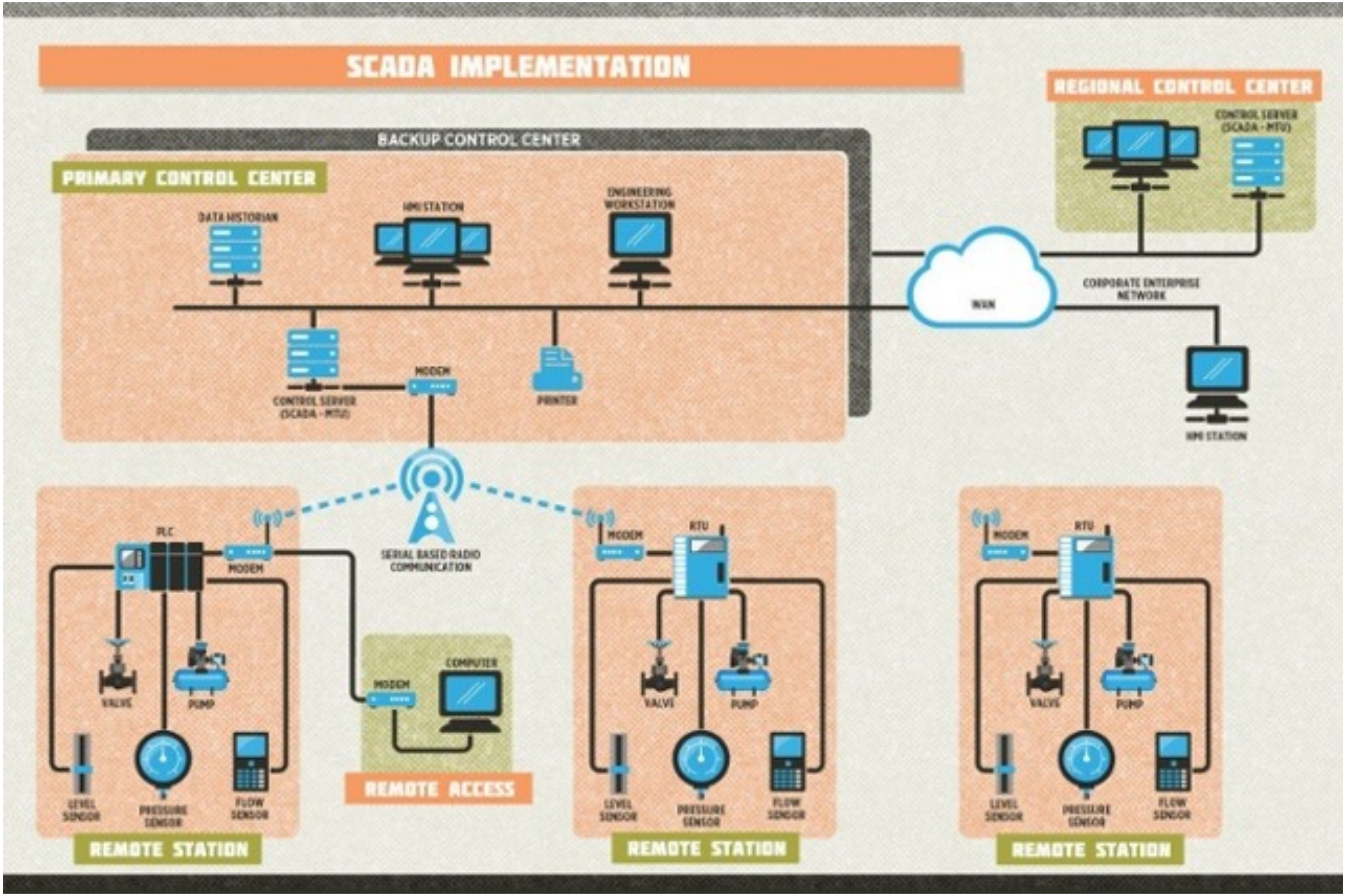
[VC18]

SCADA SYSTEMS

- **SCADA** stands for Supervisory Control and Data Acquisition
  - reference Machine to Machine (M2M) systems used in industries since mid 1970
  - supervisory systems to exert **remote control on devices/ systems/plant**
- It's a kind of **Industrial Control System**
  - i.e. computer-based systems that monitor and control industrial processes in some physical environment
  - SCADA system are large-scale ICS
    - multiple nodes, distributed over large-scale territories

SCADA SYSTEM

nice video:
https://www.youtube.com/watch?v=WQWJzgbdq1E



http://blog.cimation.com/blog/key-differences-between-scada-dcs-and-hmi-systems

SCADA: MAIN ELEMENTS - RTUs

- **Sensors and Remote terminal units (RTUs)**
  - at the base layer of a SCADA there is a sensor network
    - allow for monitoring the functioning and state of industrial machines and processes
  - connected to sensors we have the remote terminal unit (RTU)
    - their job is to collect relevant data from sensors and transform them into digital signals to be sent to the remote control station

SCADA: MAIN ELEMENTS - THE PLC

- **Programmable logic controller (PLCs)**
  - The programmable logic controller o PLC (“Controller logico programmabile”) are computers connected to the sensor network & RTU
  - PLC job is to supervise and coordinate data collection, instructing/controlling both RTU and sensors
    - it is the PLC that that has the “super-loop”, specifying when to do the sampling
  - programming languages: IEC 61131-3 standard family

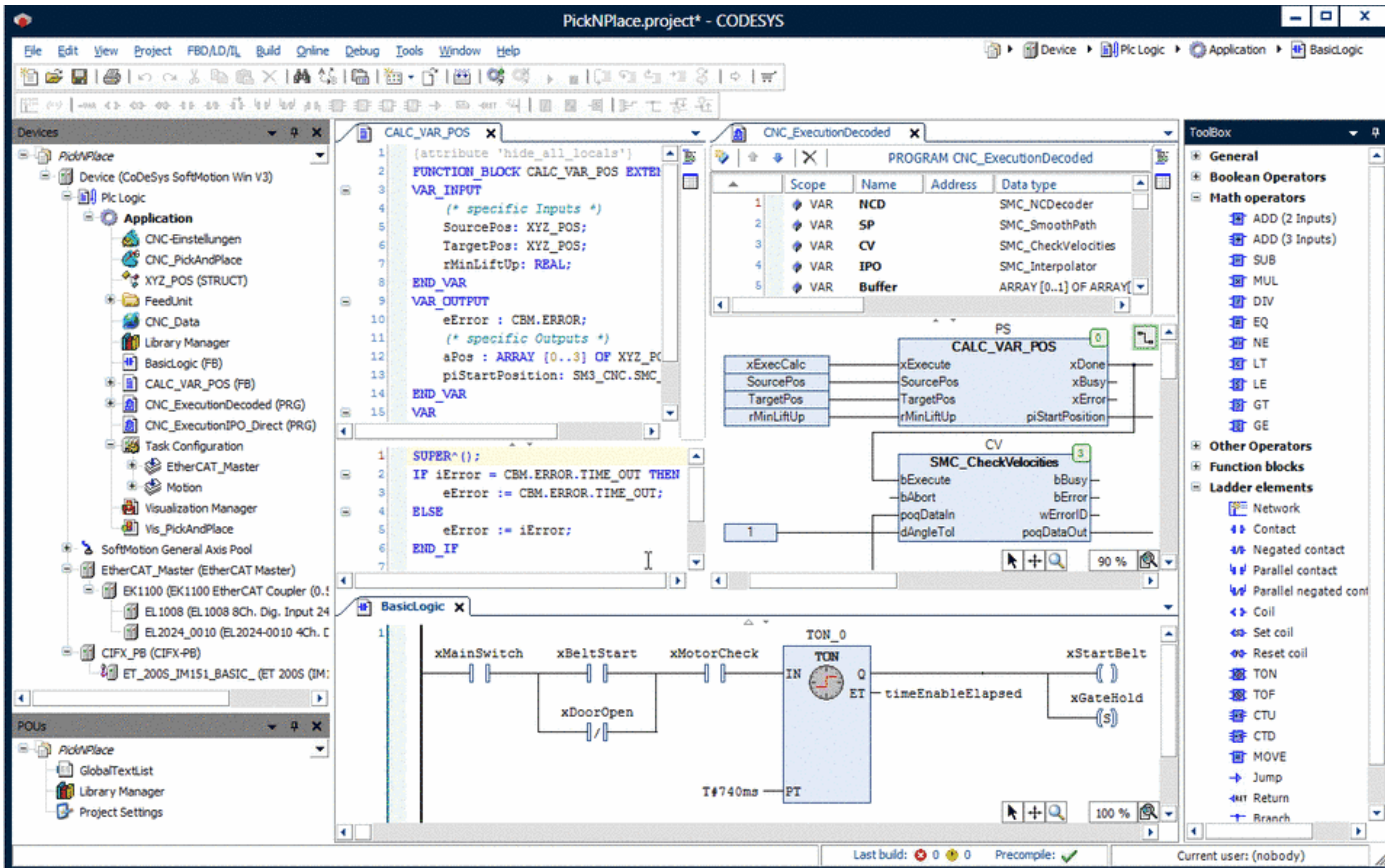
nice videos:
https://www.youtube.com/watch?v=PbAGl_mv5XI

PLC PROGRAMMING WITH IEC 61131-3

- IEC 61131-3 is the third part of the international standard IEC 61131 for PLC (programmable logic controllers).
- It defines a set of programming languages:
  - **Ladder diagram (LD)**, visual
  - **Function block diagram (FBD)**, visual
  - **Structured text (ST)**, textual
  - **Instruction list (IL)**, textual
  - **Sequential function chart (SFC)**
    - this is the main one, used to realise complex control programs

nice video:
https://www.youtube.com/watch?v=Qf32qtHfowQ

PLC PROGRAMMING WITH IEC 61131-3



SCADA: OTHER ELEMENTS

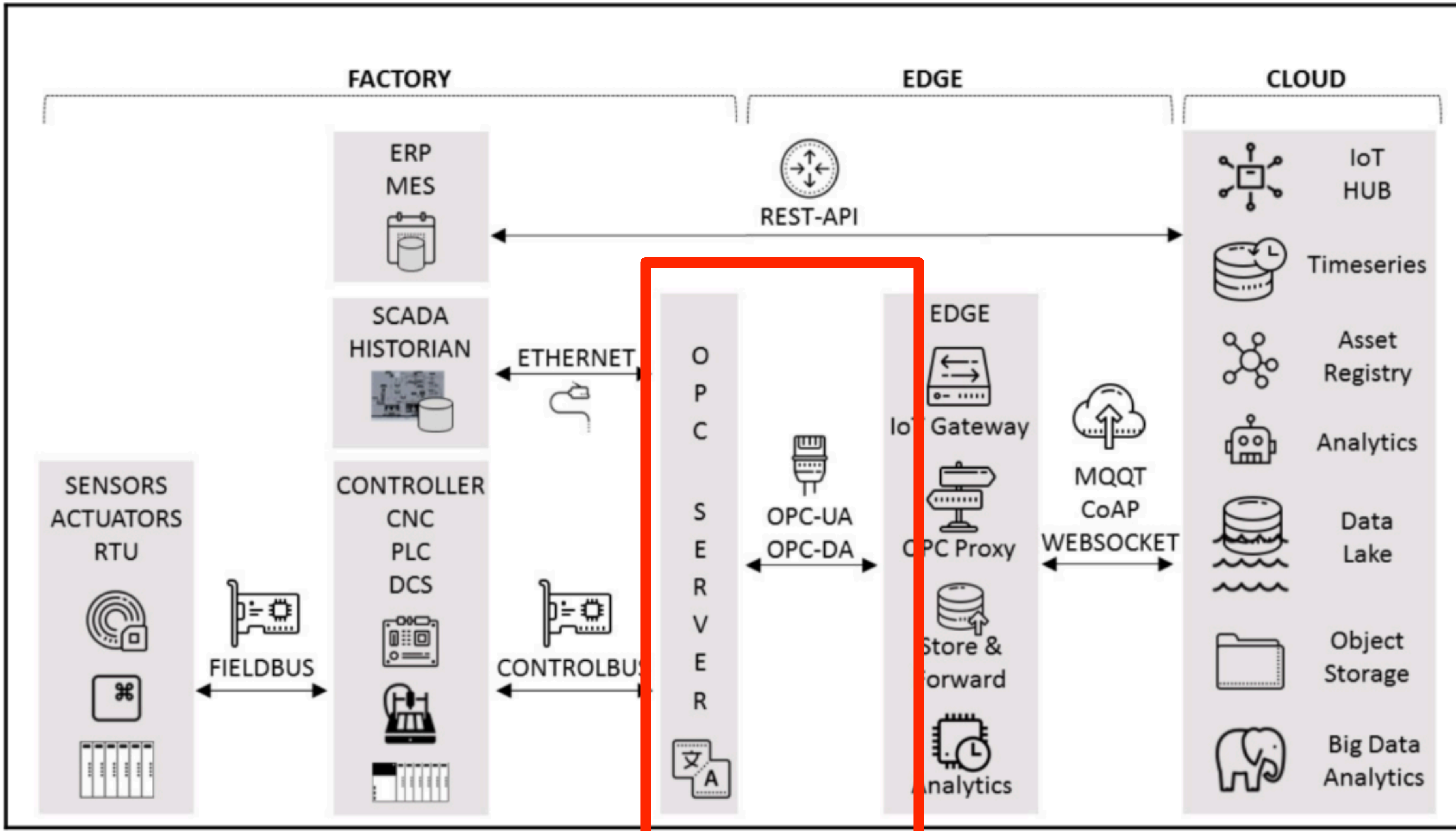
- **Telemetry system**, to connect PLC and RTU to central control station, data warehouse, and the enterprise system
  - can be a computer network such as a LAN o even a WAN, or it can be ad-hoc serial line
- **MTU (Master Terminal Unit)**
  - control server that periodically collect all data from PLC, process the data to get useful info, store the info and send control commands
- **HMI (human machine interface)**
  - interface that allows human operators to access to data, monitor them and interact with the system
- **Historian**
  - software service storing time-stamped data, events, alarms — so a database which is queried for creating graphs and reports on the HMI

SCADA HMI - EXAMPLE

http://www.armani-engr.com/hmiscada.html

I-IOT DATA FLOW

- From its generation at the field level to its processing in the cloud



THE OPC STANDARD

- **OPC** is the acronym of **Open Platform Communication**
  - software layer that makes it possible to interact with the industrial data source through its own protocol for querying tags and time-series, and exposing them by means of a *standard interface* to the upper levels and external systems
  - acting as a *translator* between the controllers and producers of the industrial data and the consumer systems
    - used to abstract PLC or DCSes specific protocols (e.g. Modbus and Profibus) into a standardized interface
    - function as a middle-man for SCADA & co converting generic OPC read & write requests into device-specific requests and vice-versa
- Through OPC, PLCs or DCSes *can expose tags, time-series, alarms, events and Sequence of Events (SOE) to any system that implement the OPC interface*
  - SCADA and Historian where initially developed as consumers of industrial data, implementing an *OPC client* interface, but themselves become producers of industrial data, implementing an *OPC server* interface

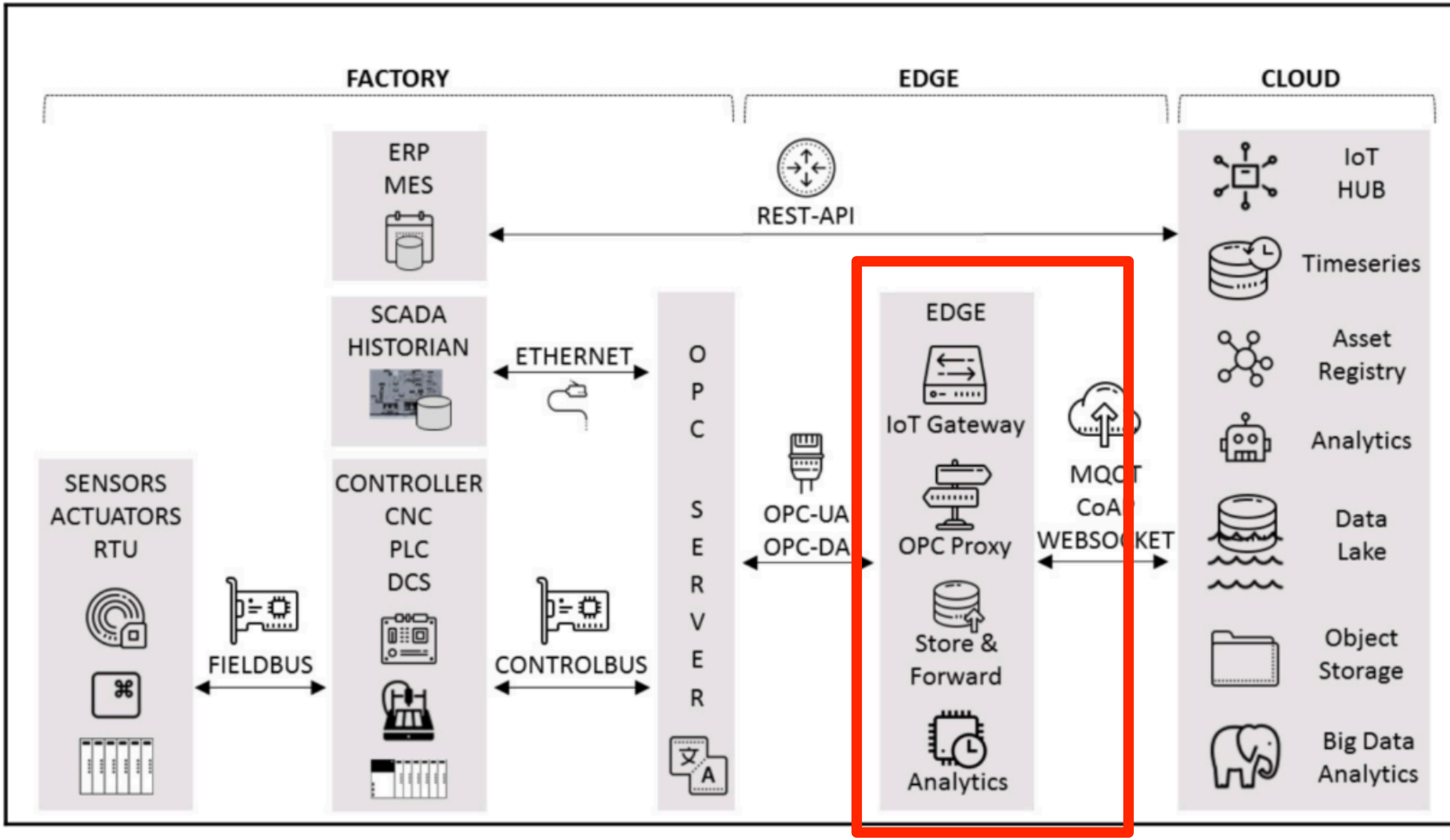
OPC SPECIFICATIONS

- Two main specifications
  - **OPC Classic** (old, ~1995)
    - based on Windows OS and OLE / DCOM
  - **OPC-UA** (~2005)
    - based on service-oriented approach as defined by IEC 62451
    - not only for Windows, to be embeddable also in small devices

nice intro/overviews:
https://www.youtube.com/watch?v=-tDGzwsBokY
https://www.youtube.com/watch?v=vRk42W_4R0o

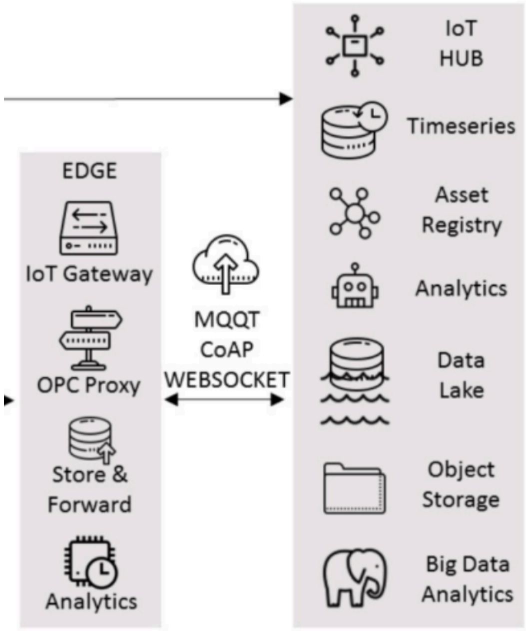
I-IOT DATA FLOW

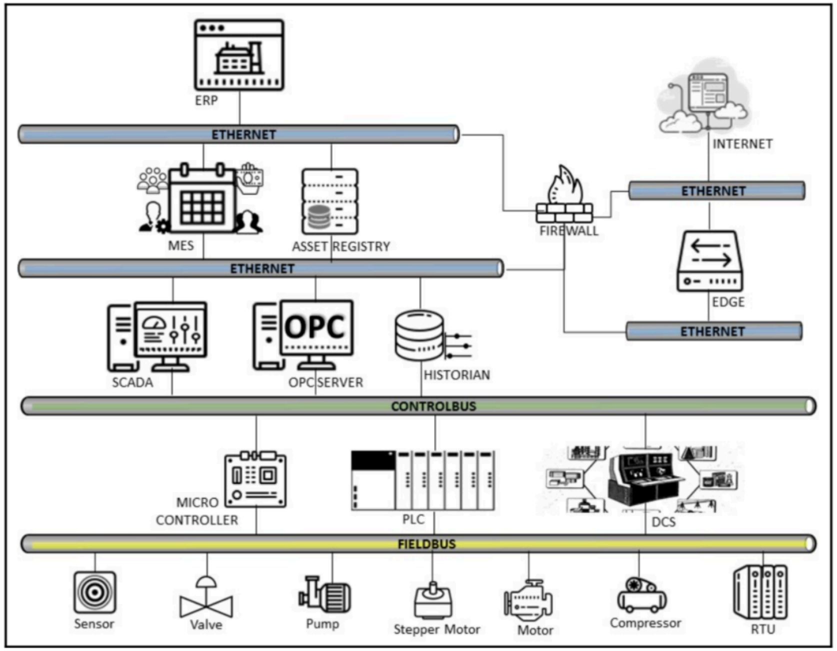
- From its generation at the field level to its processing in the cloud



THE EDGE DEVICE

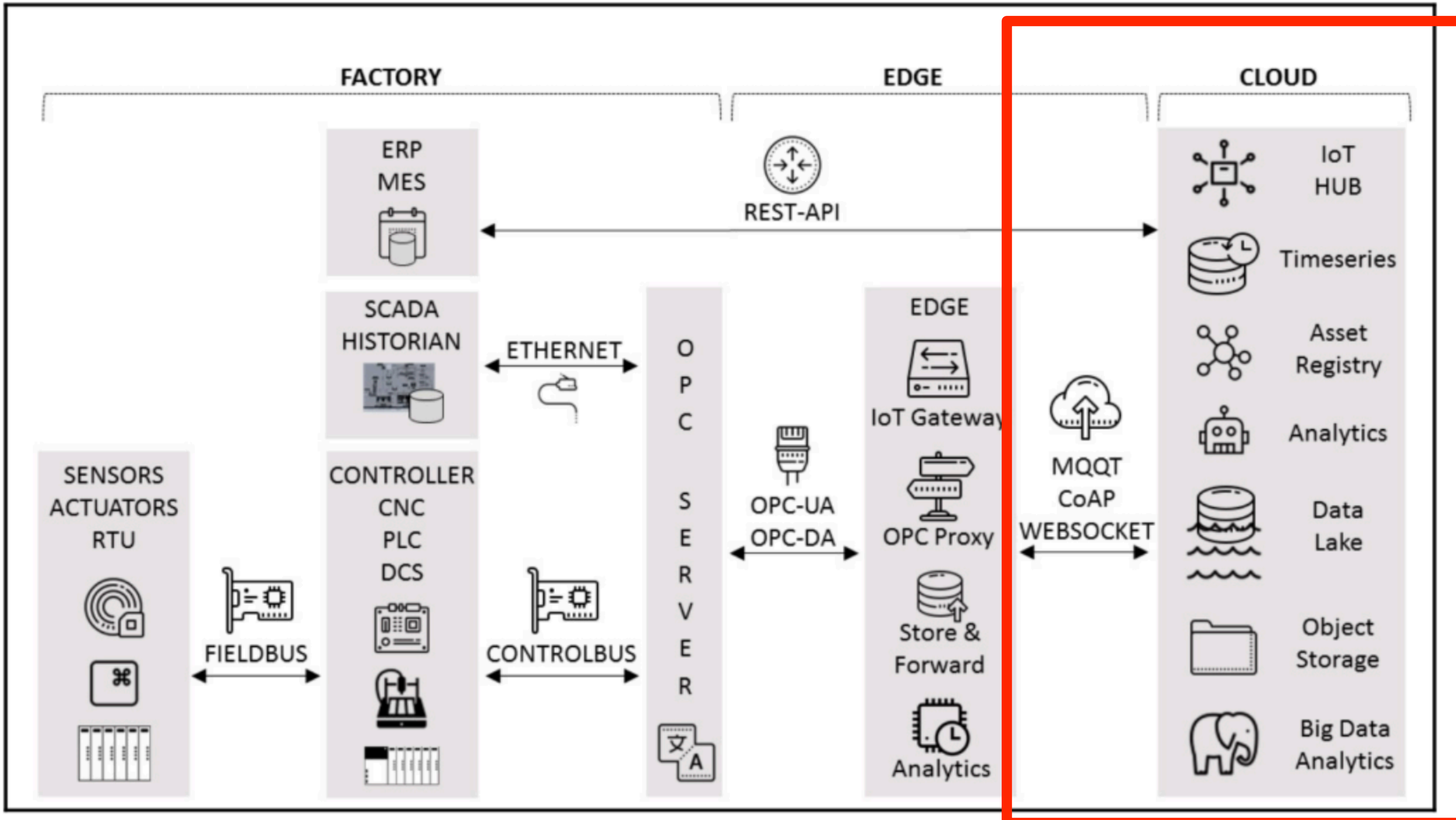
- **Device which enables the IIOT**
  - physically located in the factory
  - linked from one side to the industrial data sources through the OPC server, and from the other side to the cloud through the IoT Gateway
- Different kinds of approaches in industries about arranging and conceiving the edge device





I-IOT DATA FLOW

- From generation to processing in the cloud



[VC18]

IOT E CLOUD

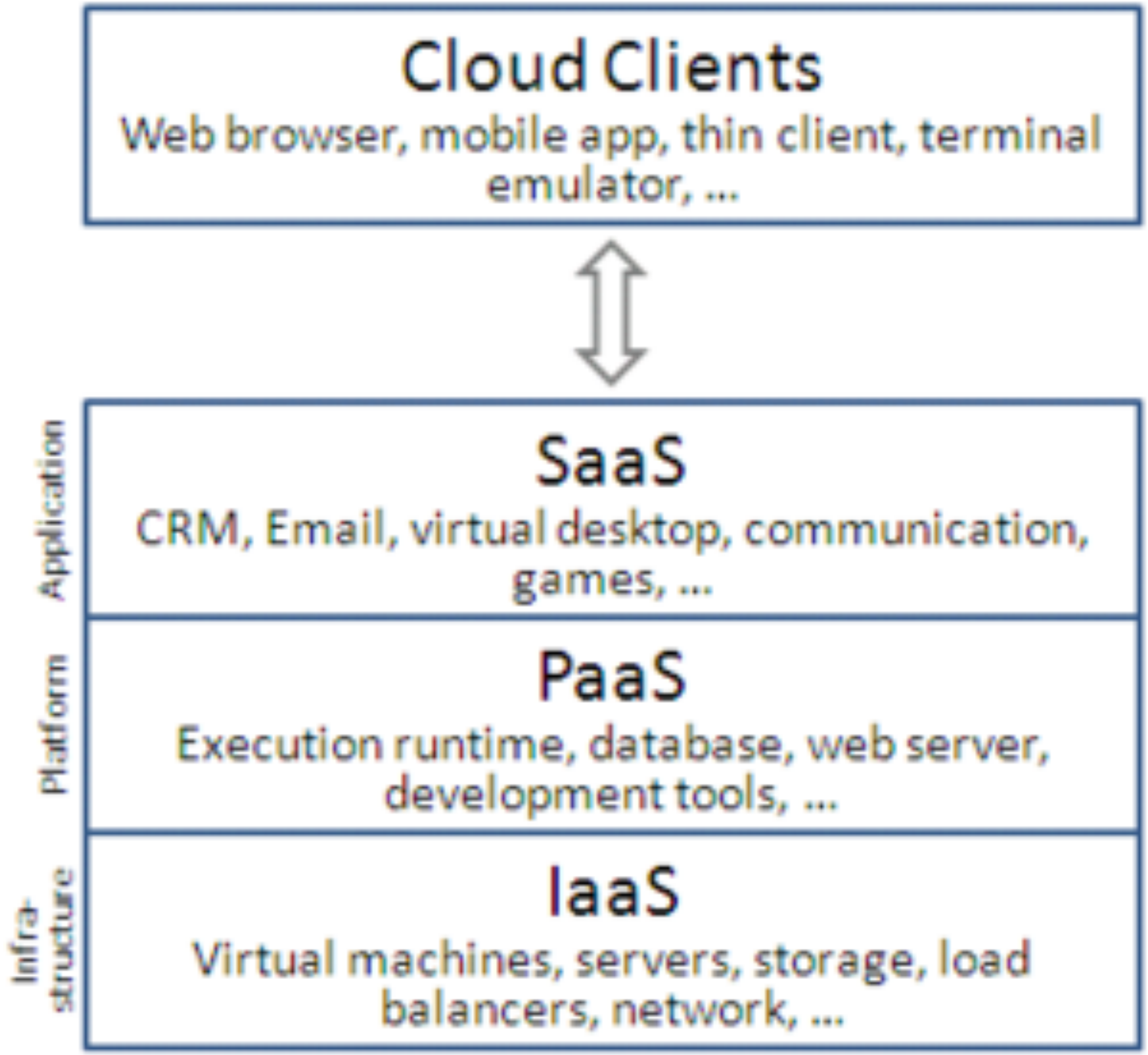
- Key feature of IoT: capability to directly/indirectly send data to the *Internet*
- This allows for managing big data and big streams generated by sensors
  - that could not be stored locally
- The info become accessible by means of specific Internet or Web services, enabling the exchange with other applications, in particular mobile app
- Important role of the **cloud**
  - services fo device management, data storage, offline analysis, etc.
  - open systems

CLOUD

- "Cloud computing"
  - *Cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction.* (*)
- Resources are provisioned **as-a-service** on the network and can be at three different levels:
  - **IAAS** - infrastructure as a service
    - virtualised HW / resources — servers, storage, network
  - **PAAS** - platform as a service
    - execution runtime, database, web servers, IDEs,…
  - **SAAS** - software as a service
    - applications — CRM, office apps, mail apps, etc.
- Client side (desktop, mobile, wearable, embedded..)
  - modern browsers - HTML5, JavaScript

(*) Peter Mell, Timothy Grance, The NIST Definition of Cloud Computing. NIST, Special Publication 800-145, Settembre 2011.

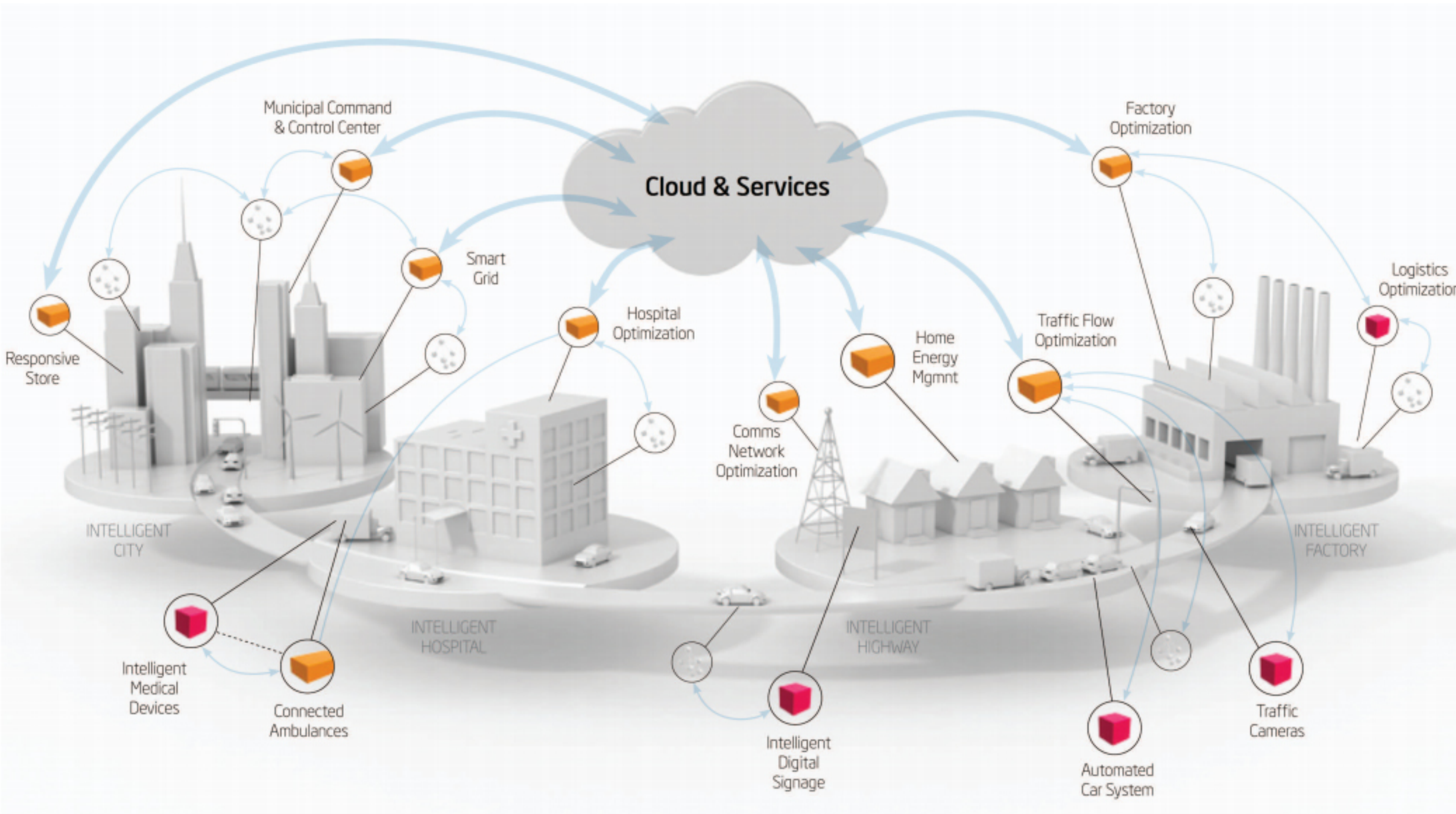
CLOUD LEVELS



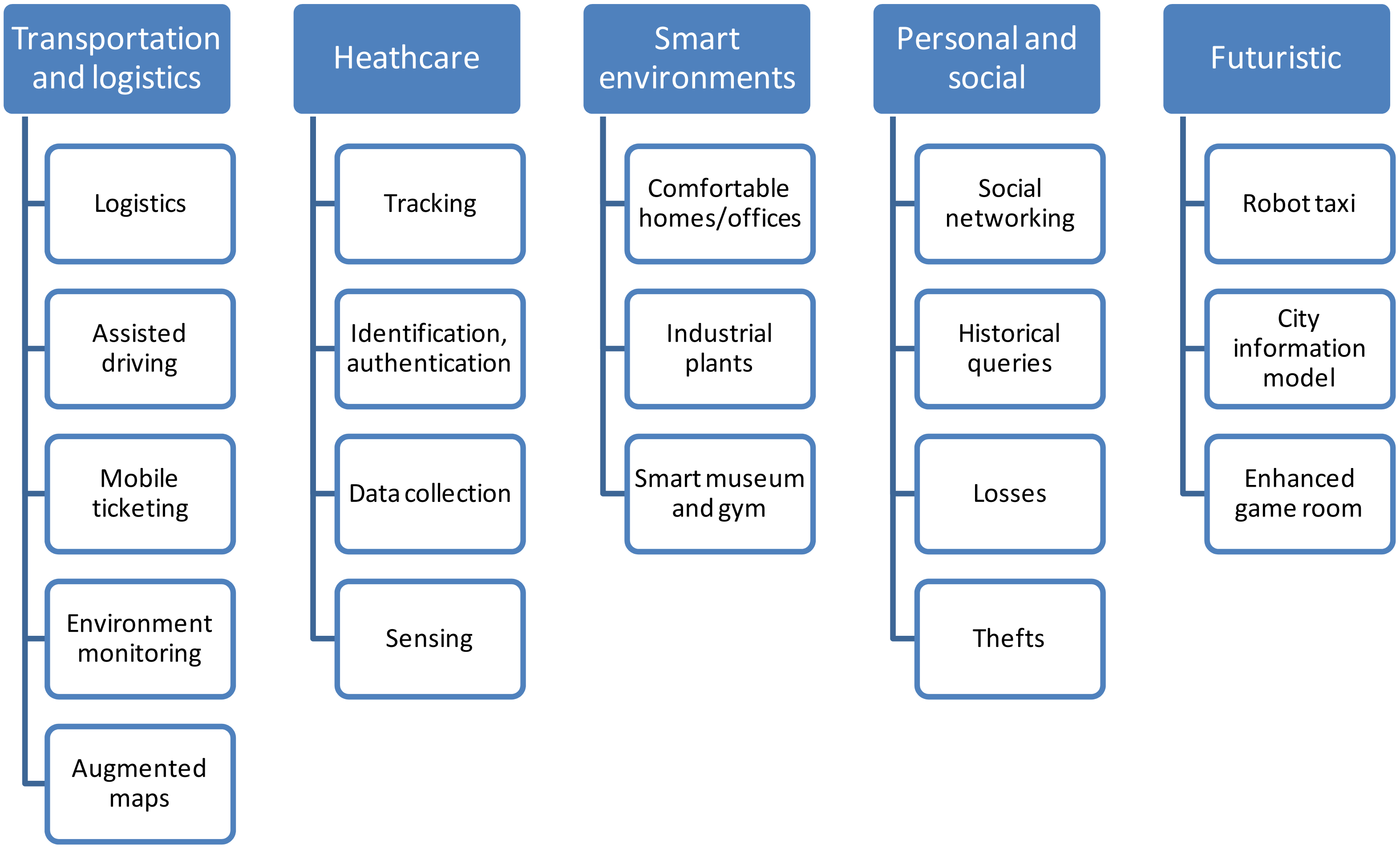
CLOUD AND IOT

- In the IoT case, cloud services are primarily used for:
  - collecting data from devices
    - stream
  - accessing to the collected data
    - pull, event-driven
    - offline and online analysis
  - device management
- **Big Data**
  - the volume and velocity of the generated data could be large, calling for specific techniques for storing, access, and manage

IOT AND THE CLOUD

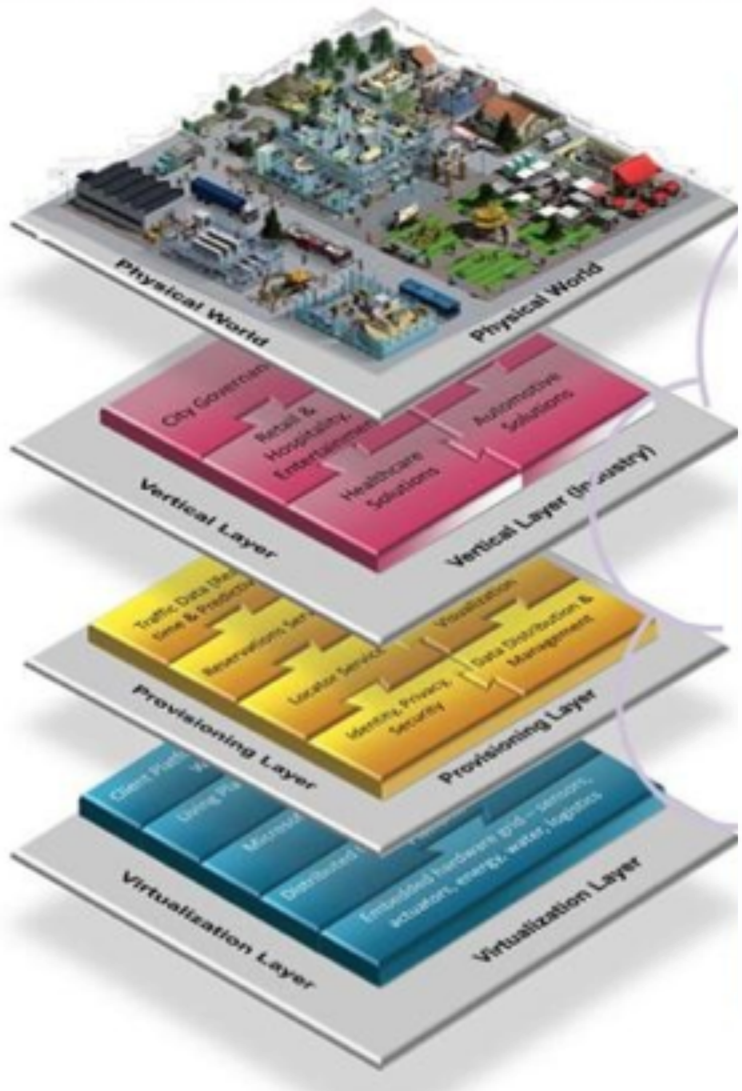


IOT APPLICATION DOMAINS



Tratto da: [Luigi Atzori, Antonio Iera, Giacomo Morabito.
The Internet of Things: A survey. Computer Networks, 54 (2010)]

SMART CITY, URBAN OS & APPS



(Urban OS, Plan IT)

INTERNET OF THINGS
CLOUD

Examples: Future Scenarios

Weather

Building and vehicle sensors provide microclimate information enabling improved models; transportation and HVAC systems take current local weather into account

Emergency Services

Improved emergency planning and response using real time information and signaling

Sustainability & Utilities

Smart grid, water efficiency, energy demand and supply management, shaping, and remote control

Transportation

Congestion avoidance, green routes, integrated systems of buses, trains, taxis, cars (with parking/charging), bikes, walking, and PRT

Vehicles

Onboard energy generation and/or storage

IOT: CRITICAL ASPECTS AND CHALLENGES

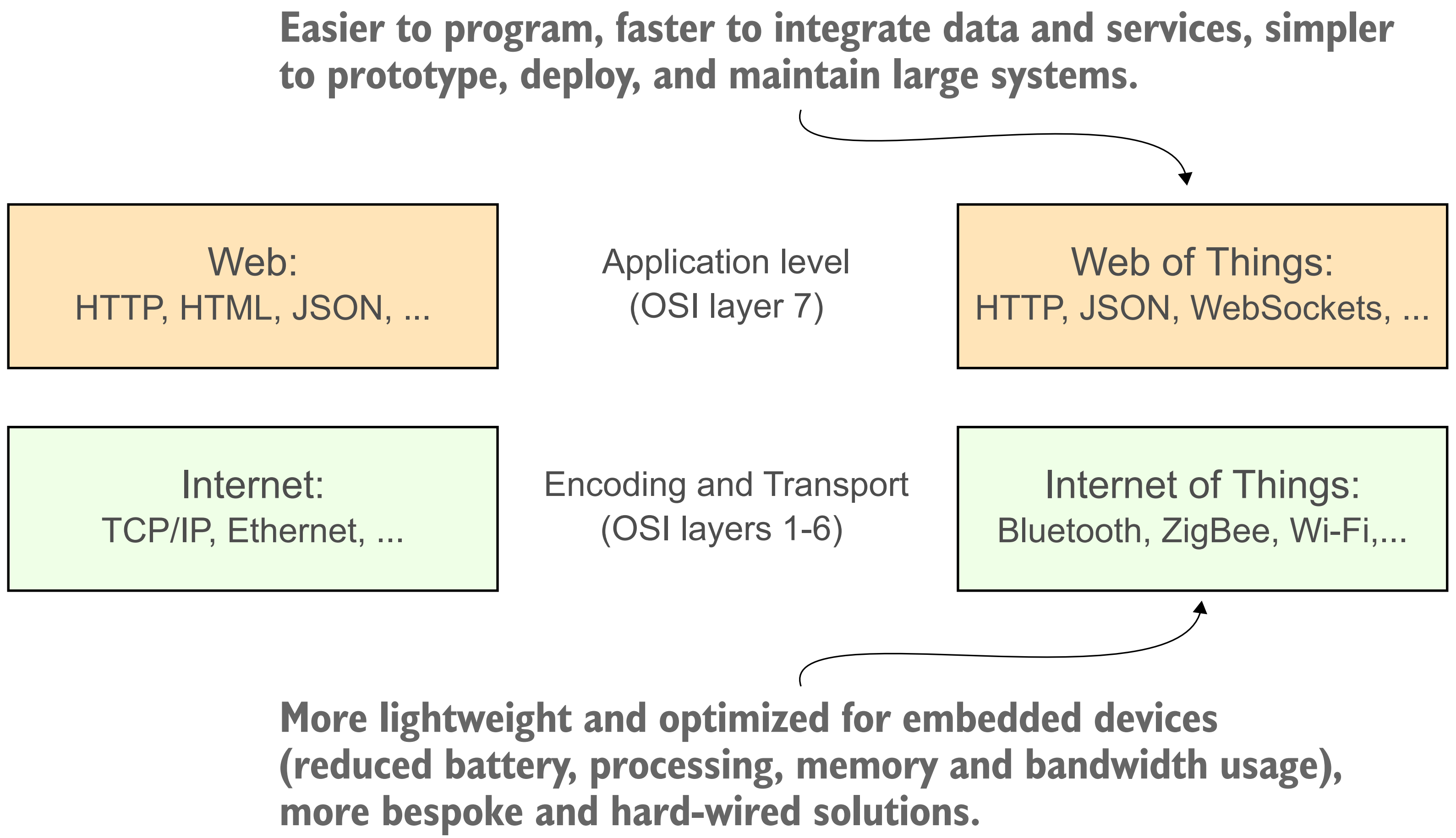
- Security
- Privacy
- Data ownerships
- Application-level interoperability
  – definition of standards and ref. architectures
  – e.g. **Web of Things** (WoT) initiative

IoT + WEB = *WEB-OF-THINGS*

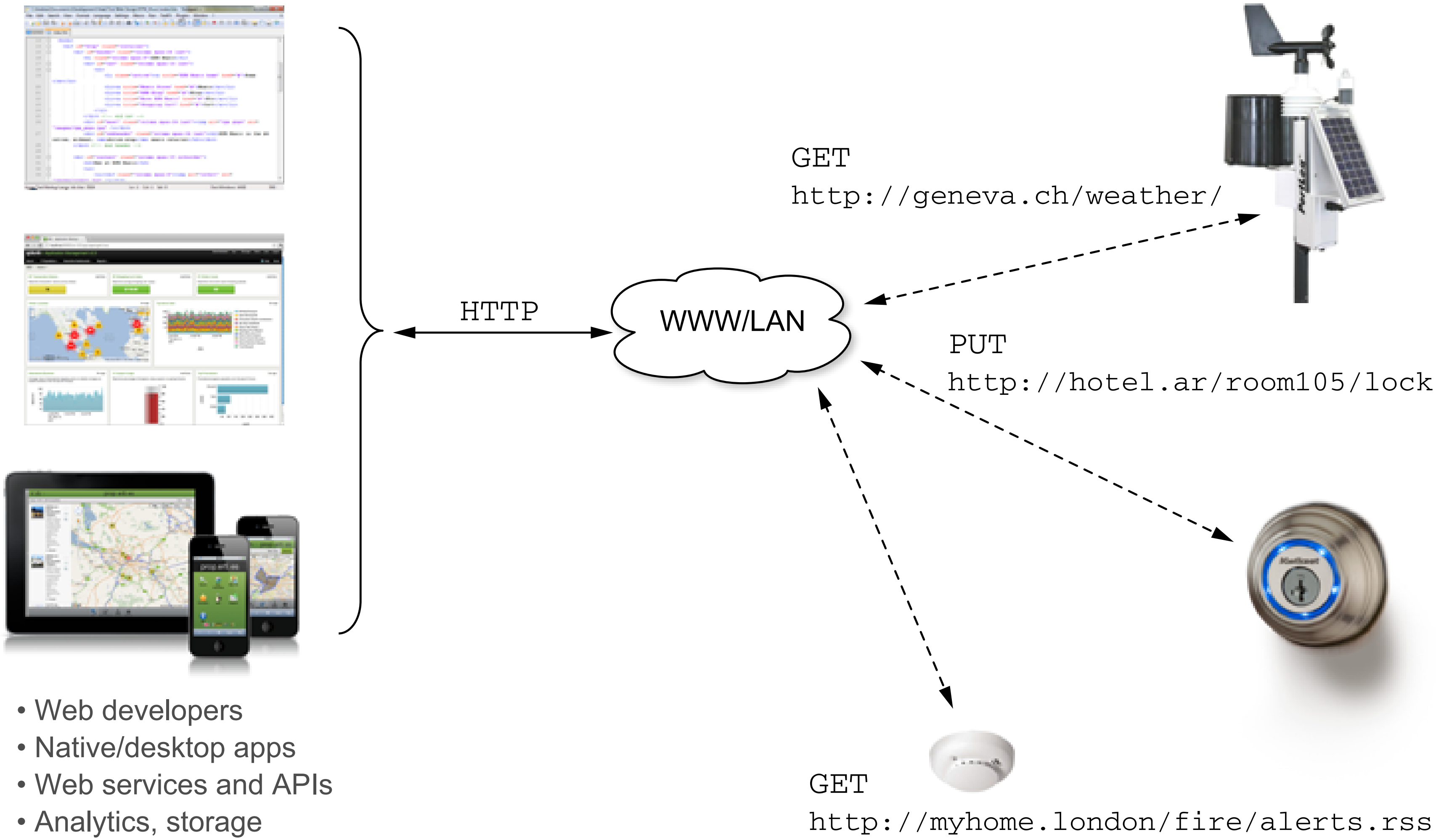
- **Web of Things** (o **WoT**) bringing IoT things into the **World Wide Web,** providing each thing a REST-ful API
  - enabling interoperability *at the application level*
  - enabling interaction with the thing using the web, in spite of the lower level protocol used thing implementation
    - Zigbee, Bluetooth, 6LoWPAN…

Guinard, Dominique; Vlad Trifa; Erik Wilde (2010). "A Resource Oriented Architecture for the Web of Things". Proc. of IoT 2010 (IEEE International Conference on the Internet of Things). Tokyo, Japan.

WoT = IoT + APPLICATION LEVEL

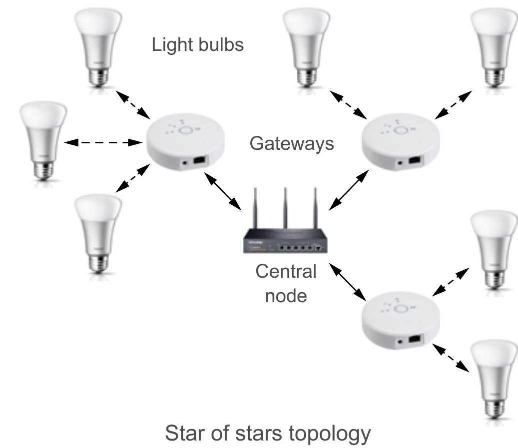


WEB OF THINGS (WoT)

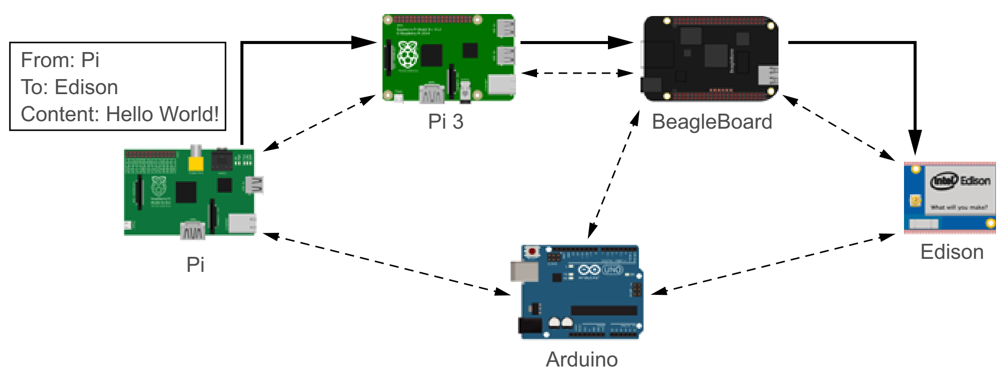


WOT - NETWORK TOPOLOGIES

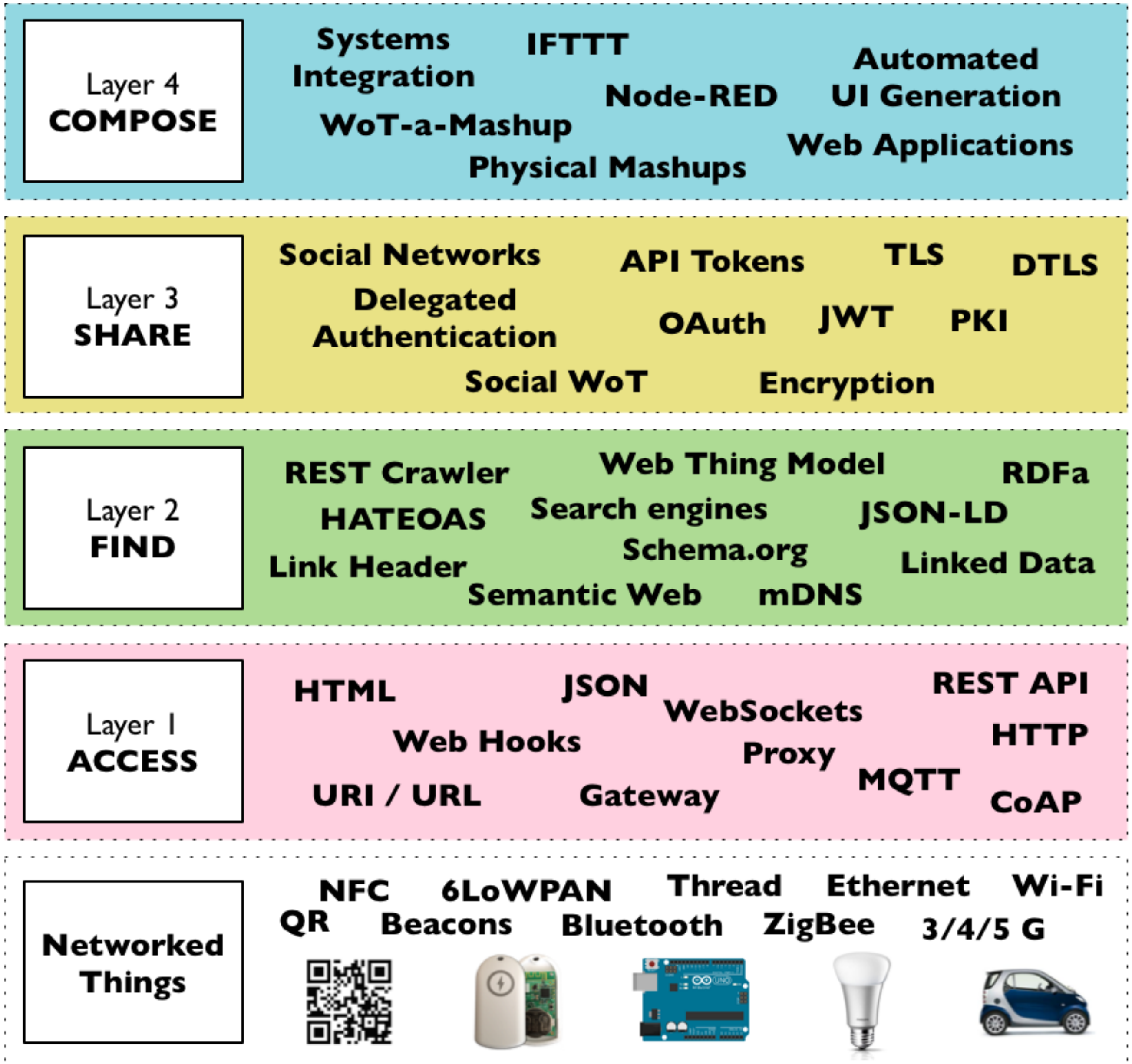
- Star topologies



- Mesh topologies
  - mesh networks
  - relays



WOT LEVELS

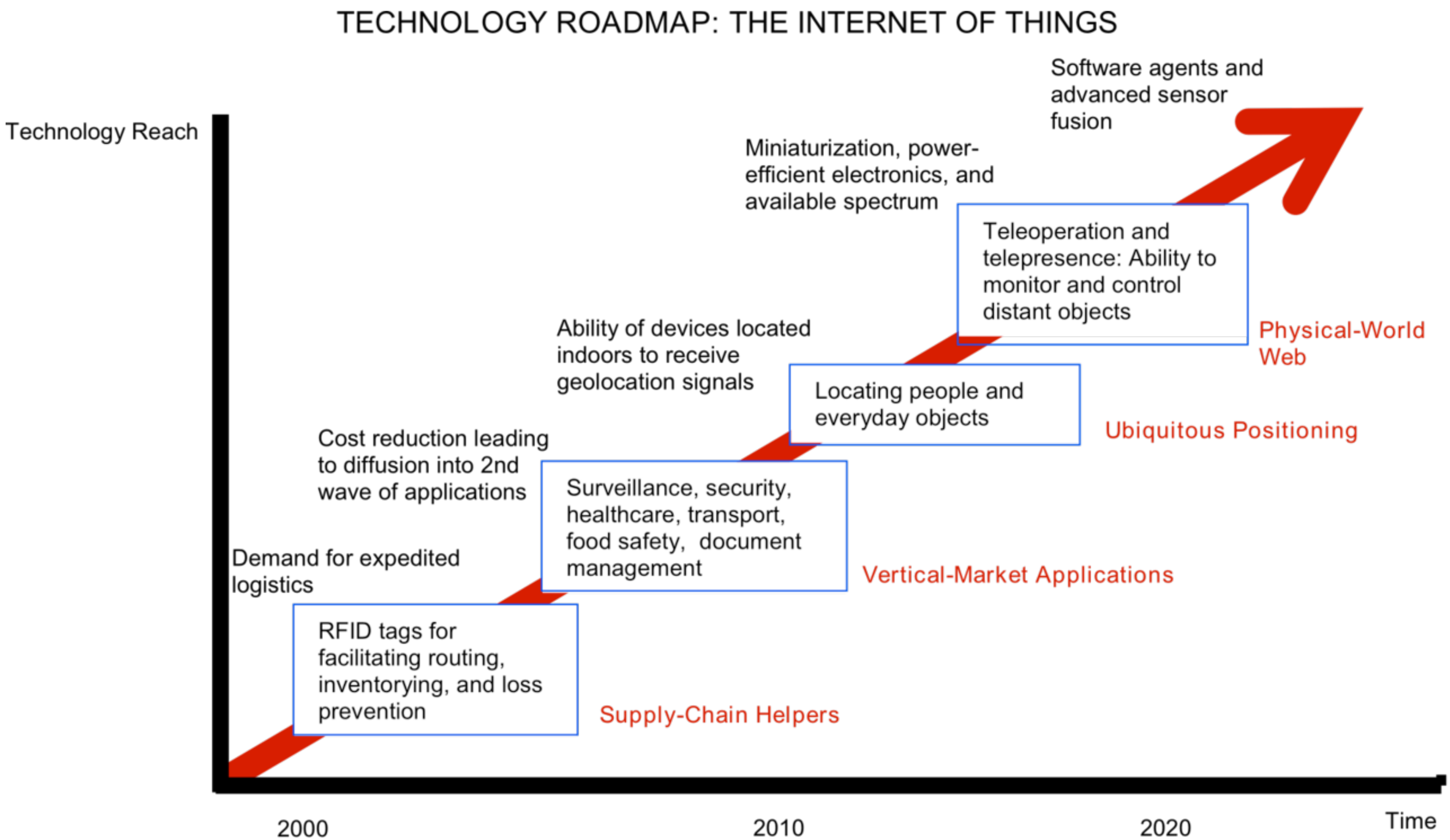


Source: Building the Web of Things: book.webofthings.io

WEB OF THINGS AT W3C

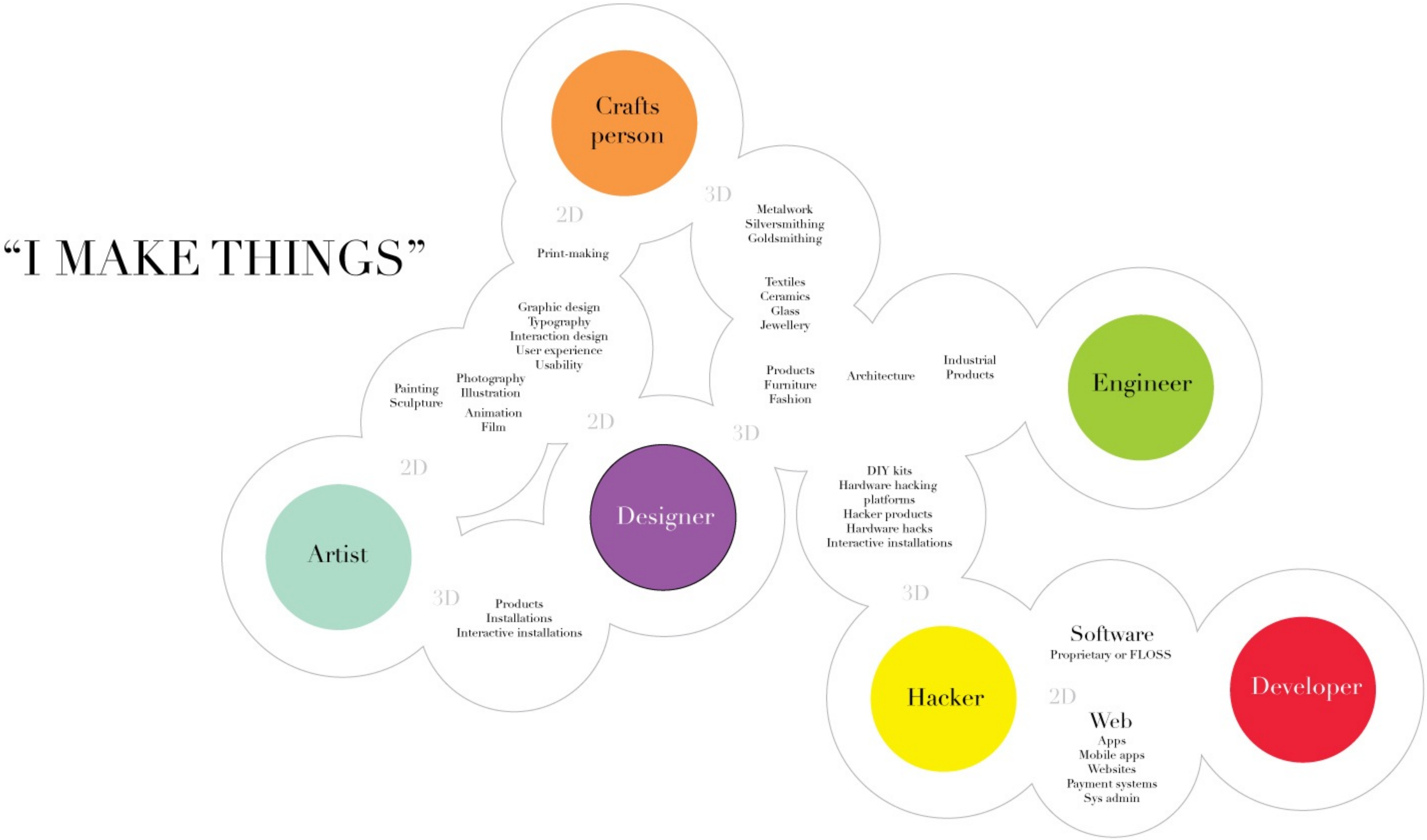
- Web of Things Interest Group
  - https://www.w3.org/WoT/
- Web of Things Working group
  - https://www.w3.org/WoT/WG/
- First Submission of the Web Thing Model
  - https://www.w3.org/Submission/2015/01/
- Current proposals
  - WoT Architecture
    - https://w3c.github.io/wot-architecture/
  - WoT Things Description
    - https://www.w3.org/TR/wot-thing-description/

IOT: TECHNOLOGY ROADMAP



Source: SRI Consulting Business Intelligence

WHO IS MAKING IOT: MAKERS AND TINKERERS



MOBILE COMPUTING AND IOT

- **Mobile computing**
  - software systems running on mobile devices
    - smartphones, tablets
  - peculiar aspects
    - user interaction
    - energy-awareness
    - intermittent network connection
    - location-based/***context-aware*** app
    - …

MOBILE COMPUTING AND IOT

- In IoT two main aspects:
  - mobile devices as interface to interact with smart Things & Environments
    - Personal Area Network (PAN) - Bluetooth, BLE, NFC.. - + Internet
  - mobile devices as IoT Things themselves
    - equipped with many sensors, actuators, network connection
    - tracking user-related data

WEARABLE COMPUTING AND IOT

- **Wearable computing** & **eyewear computing**
  - wearable devices
    - smart-watch, smartglasses,…
    - typically coupled with mobile devices
      - Bluetooth
  - peculiar aspect
    - *hands-free* / *on-the-go*
- IoT context
  - interface for extending the interaction with physical environments ***augmented,*** extended with a virtual functionalities
  - Augmented Reality
    - ..Mixed Reality, Extended Reality

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